Section 4

Thursday, February 5, 2026 8:16 PM

- <u>Transformation of Random variables</u>

X is a random variable w/ distribution $f(x)$ and moment generating function $M_X(t)$.

Let $Y = g(X)$, what is the distribution of Y?

1) Use $f(x)$ to find distribution of Y

Step 1) find $g^{-1}(y)$

$$Y = g(X) \rightarrow X = g^{-1}(y)$$

Step 2) Calculate the "Jacobian"

$$\left|\frac{dx}{dy}\right| = \left|\frac{dg^{-1}(y)}{dy}\right|$$

Step 3) Find the pdf of y

$$h(y) = f(g^{-1}(y)) \cdot \left|\frac{d}{dy} g^{-1}(y)\right|$$

Step 4) determine support of y

2) Use $M_Y(t)$ to find distribution of Y

$$M_Y(t) = E(e^{tY}) = E(e^{tg(x)}) \cdots$$

- Common Types of Transformations:

$X \sim f(x)$ $\quad x > 0$

$y = \log(x)$ $\quad -\infty \le y \le +\infty$

$x = e^y$ $\quad x > 0$

$X \sim f(x)$

$Y = X^k$

$Y = aX + b$ (scale and shift)

go over this first...

- <u>Sum of independent Random Variables</u>

$X_1, X_2, \ldots, X_n \overset{iid}{\sim} N(\mu, \sigma^2)$ $\quad -\infty \le x_i \le +\infty$

$Y = \sum_{i=1}^{n} X_i$, what is the distribution of Y?

★ note: $M_{X_i}(t) = E(e^{tX_i}) = \exp\left\{\mu t + \frac{1}{2}\sigma^2 t^2\right\}$

$$M_Y(t) = E(e^{tY}) = E(e^{t\sum_{i=1}^{n} X_i}) = E(e^{tX_1} \cdot e^{tX_2} \cdots e^{tX_n})$$

$$= E(e^{tX_1}) \cdots E(e^{tX_n}) = [E(e^{tX_1})]^n = \exp\left\{n\mu t + \frac{n}{2}\sigma^2 t^2\right\}$$

2nd

So, $Y \sim N(n\mu, n\sigma^2)$

- now $Y = \frac{1}{n}\sum_{i=1}^{n} x_i$?

$$M_Y(t) = E(e^{tY}) = E\left(e^{\frac{t}{n}\sum_1 x_i}\right) = E\left(e^{\frac{t}{n}x_1}\cdots e^{\frac{t}{n}x_n}\right) = E\left(e^{\frac{t}{n}x_1}\right)\cdots E\left(e^{\frac{t x_n}{n}}\right)$$

$k = \left(\frac{t}{n}\right)$

$$= E(e^{kx_i})^n = \exp\left\{n\left(\mu k + \frac{1}{2}\sigma^2 k^2\right)\right\}$$

$$= \exp\left\{n\mu\left(\frac{t}{n}\right) + \frac{1}{2}\sigma^2\left(\frac{t}{n}\right)^2\right\}$$

$$= \exp\left\{\mu t + \frac{1}{2}\frac{\sigma^2}{n}t^2\right\}$$

3rd

So, $Y = \frac{1}{n}\sum x_i = \bar{X} \sim N(\mu, \sigma^2/n)$

$X_i \overset{ind}{\sim} N(\mu_i, \sigma_i^2) \quad i = 1, \ldots, n$

$Y = \sum_{i=1}^{n} x_i$

$$M_Y(t) = E(e^{tY}) = E(e^{t\sum x_i}) = E(e^{tx_1})\cdots E(e^{tx_n})$$

$$= \exp\left\{t\mu_1 + \frac{1}{2}\sigma_1^2 t^2 + \cdots t\mu_n + \frac{1}{2}\sigma_n^2 t^2\right\}$$

$$= \exp\left\{t\sum\mu_i + \frac{1}{2}t^2\sum\sigma_i^2\right\}$$

$Y \sim N(\sum\mu_i, \sum\sigma_i^2)$

first